

Analytic expressions for streaming instability growth rates with some preliminary applications

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The Aim: Develop various analytic formulae for the fastest growing SI modes ($\sigma_{\max}$) and their corresponding fastest growing radial wavenumbers ($k_{x,\max}$)

*Pure Streaming Instability (SI), inviscid, no stratification,
controlled by 3 parameters*

1. radial pressure gradient:

$$\Pi$$

2. midplane dust-gas mass
density ratios:

$$\epsilon \equiv \frac{\rho_p}{\rho_g}$$

3. Stokes number:

$$\text{St} \equiv \Omega_K t_s$$

All lengthscales by H , time by Keplerian frequency $\Omega_K = c_s/H$

Linear stability – see Youdin+Goodman 2005

- Perturb from basic streaming state, assume Fourier solutions

$$\sim e^{i k_x x + i k_z z - i \omega t}$$

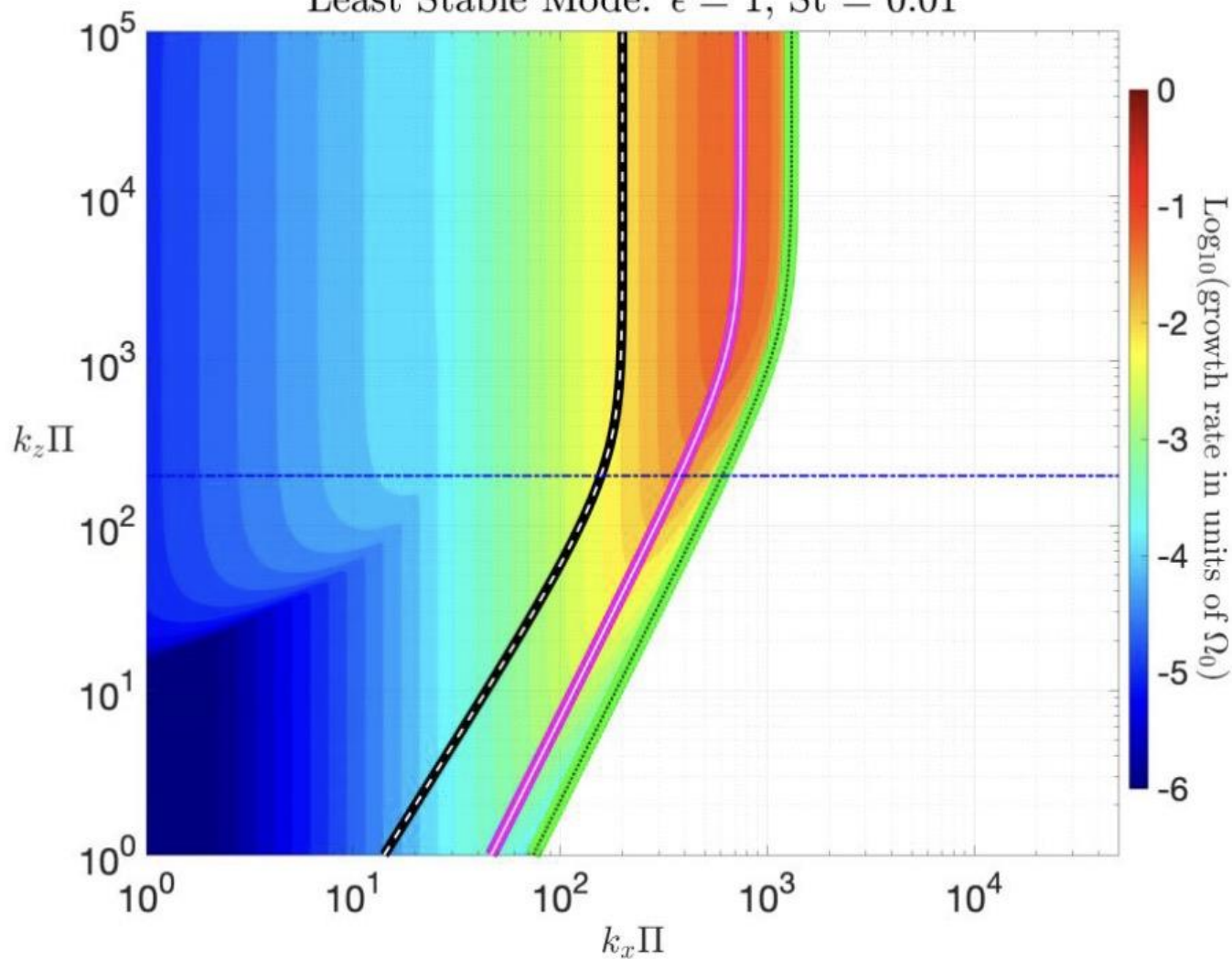
growth rate

$$\sigma \equiv \text{Im}(\omega)$$

- must solve 6th order characteristic polynomial to obtain ω ,
 - usually done using root finding methods

$$f(\omega, k_x \Pi, k_z \Pi, \text{St}, \epsilon) = 0$$

Least Stable Mode: $\epsilon = 1$, $St = 0.01$



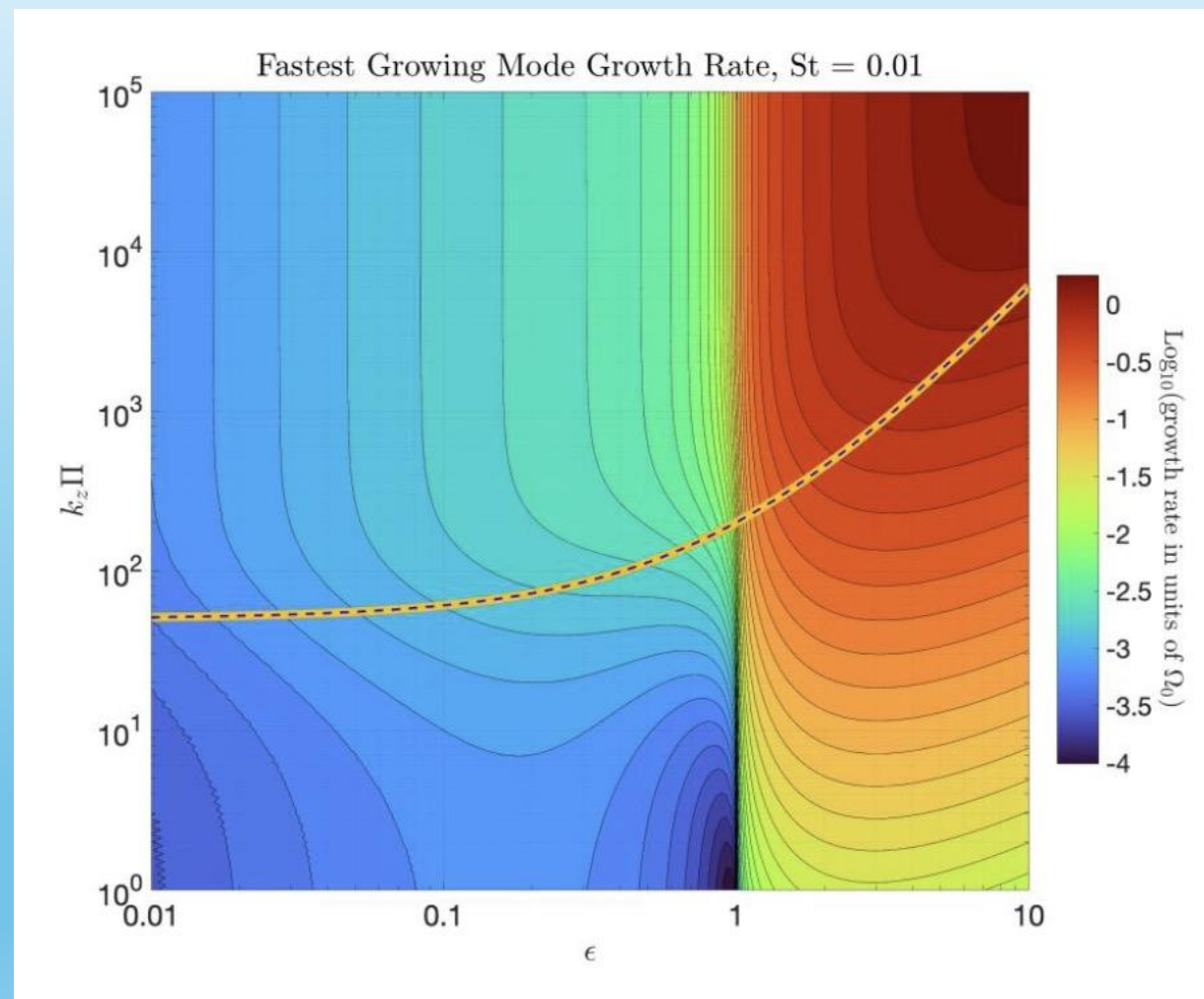
Inviscid Solutions

- All k_z unstable
- k_z given, $\exists k_{x,max}$
- k_z given, $\exists k_{x,term}$

*Identify fastest growing
modes: their σ , and k_x*

$$\sigma_{max}(k_z \Pi, St, \epsilon)$$

$$k_{x,max}(k_z \Pi, St, \epsilon)$$

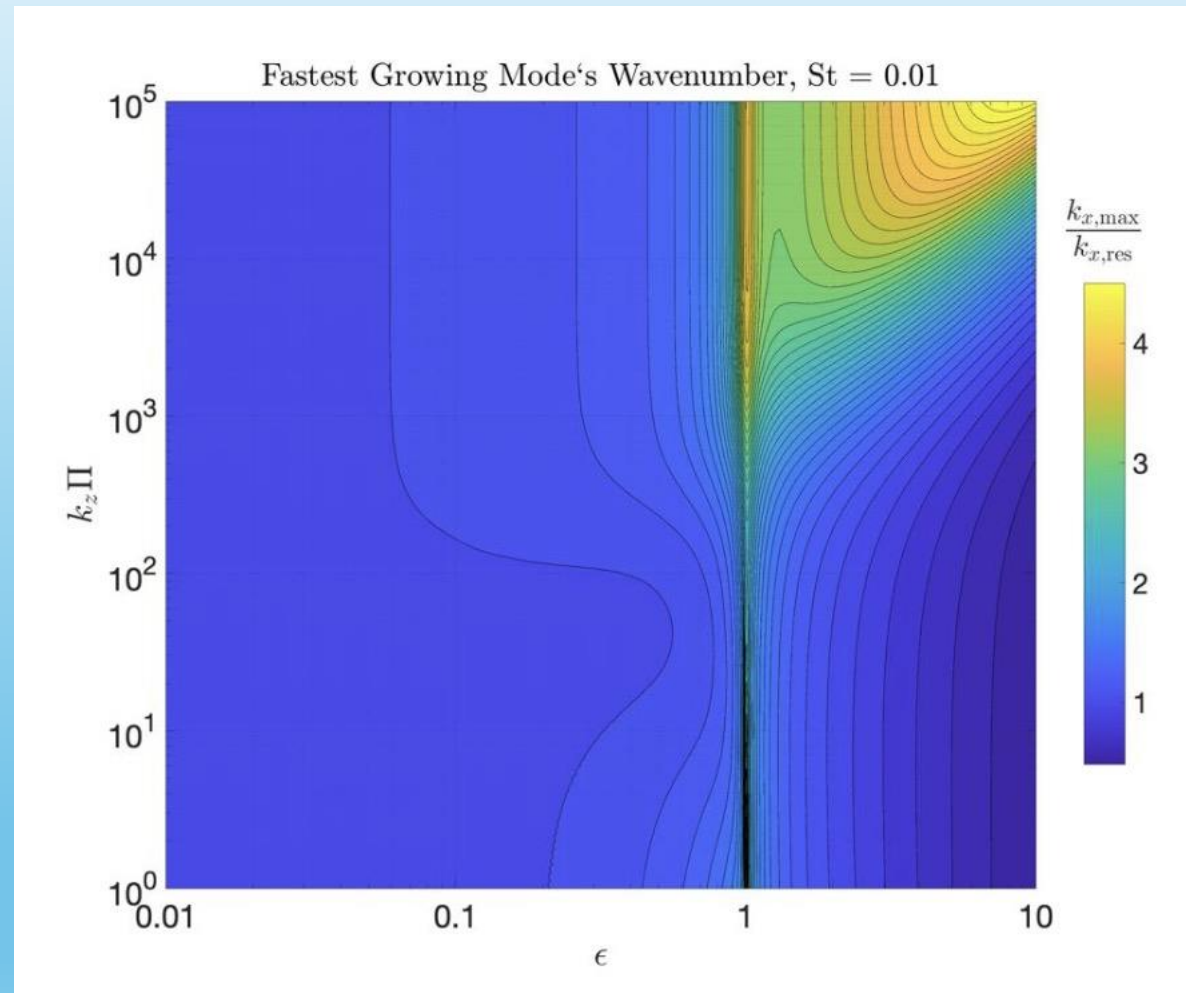


These are done via scanning solutions determined via rootfinding methods

*Identify fastest growing
modes: their σ , and k_x*

$$\sigma_{max}(k_z \Pi, St, \epsilon)$$

$$k_{x,max}(k_z \Pi, St, \epsilon)$$



These are done via scanning solutions determined via rootfinding methods

Some Previous Work on Analytic Estimates

Reduced/Model Equations

- Jacquet+11 $\sigma \propto \text{St}^3 (\epsilon - 1)^2; \text{St} \ll 1$
- Youdin+Goodman 05 $\sigma \propto \sqrt{\epsilon - 1}; \epsilon > 1$

Asymptotic Analysis, $\epsilon \ll 1$

- Squire+Hopkins 18 $\sigma \propto \sqrt{\epsilon} \cdot \Xi(\text{St}); \text{St} \text{ unrestricted}$
- Magnan+24 $\sigma \propto \sqrt{\epsilon} \cdot \text{St}; \text{St} \ll 1$

Key Insight: resonance b/w particle drift speed and inertial mode (Squire+Hopkins 2018)

- Special Wavenumber k_μ

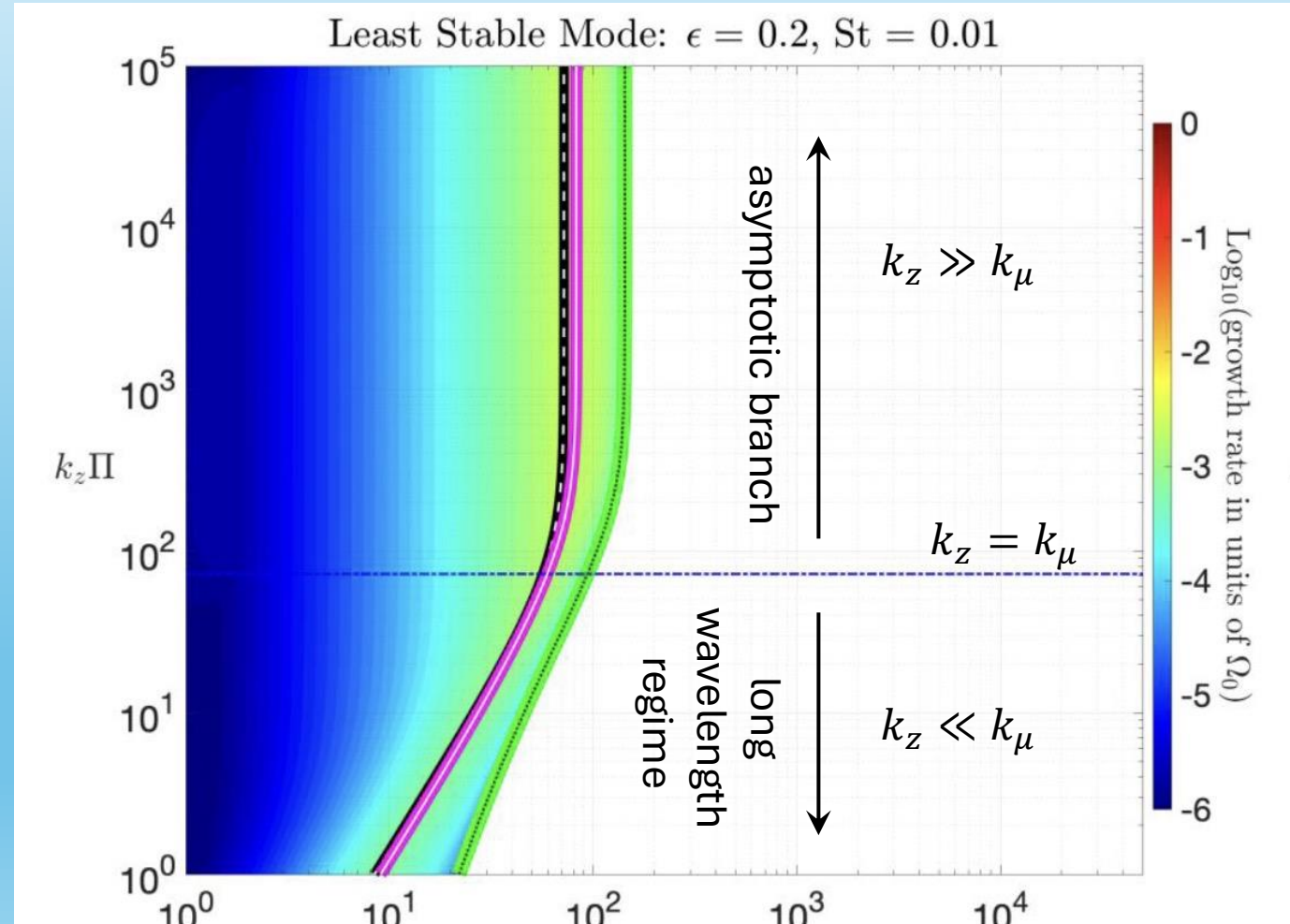
$$\Pi k_\mu \equiv \frac{(1 + \epsilon)^2}{2St}$$

- Solution Character

☐ *Asymptotic Branch*

☐ *Long wavelength Regime*

$$\Pi k_{x,res} \equiv \begin{cases} \Pi k_\mu, & k_z \ll k_\mu \\ \sqrt{k_\mu k_z \Pi}; & k_z \ll k_\mu \end{cases}$$



What do we do?

Derive asymptotically valid estimates for
 σ_{max} for $St \ll 1$ valid for any ϵ

How do we do it?

Analyze solutions to characteristic polynomial
in a power series expansions of St

Asymptotic Branch Results

$$k_z \gg k_\mu$$

$$(ii) \quad \underline{\epsilon \ll 1; \quad 2k_z \Pi St \gg (1 + \epsilon)^2:}$$

$$\sigma_{\max} \approx \sqrt{\epsilon/2} \, St + \mathcal{O}(St^2),$$
$$k_{x,\max} \Pi \approx \frac{1}{2} \, St^{-1}.$$

Recovers Squire+Hopkins 18a in St small limit

$$(vi) \quad \underline{\epsilon < 1 - St^{2/3}; \quad 2k_z \Pi St \gg (1 + \epsilon)^2:}$$

$$\sigma_{\max} \approx \frac{\epsilon}{1 + \epsilon} \frac{(\epsilon + 2)\tilde{\omega}_0 + 2(1 + \tilde{\omega}_0^2)}{1 - 2\tilde{\omega}_0 - 3\tilde{\omega}_0^2} St,$$

$$k_{x,\text{res}} \approx \frac{(1 + \epsilon)^2}{2} St^{-1}$$

$$\tilde{\omega}_0 \equiv -1 + \sqrt{\frac{\epsilon}{2}}(1 + \epsilon) + (1 - \sqrt{2})\epsilon.$$

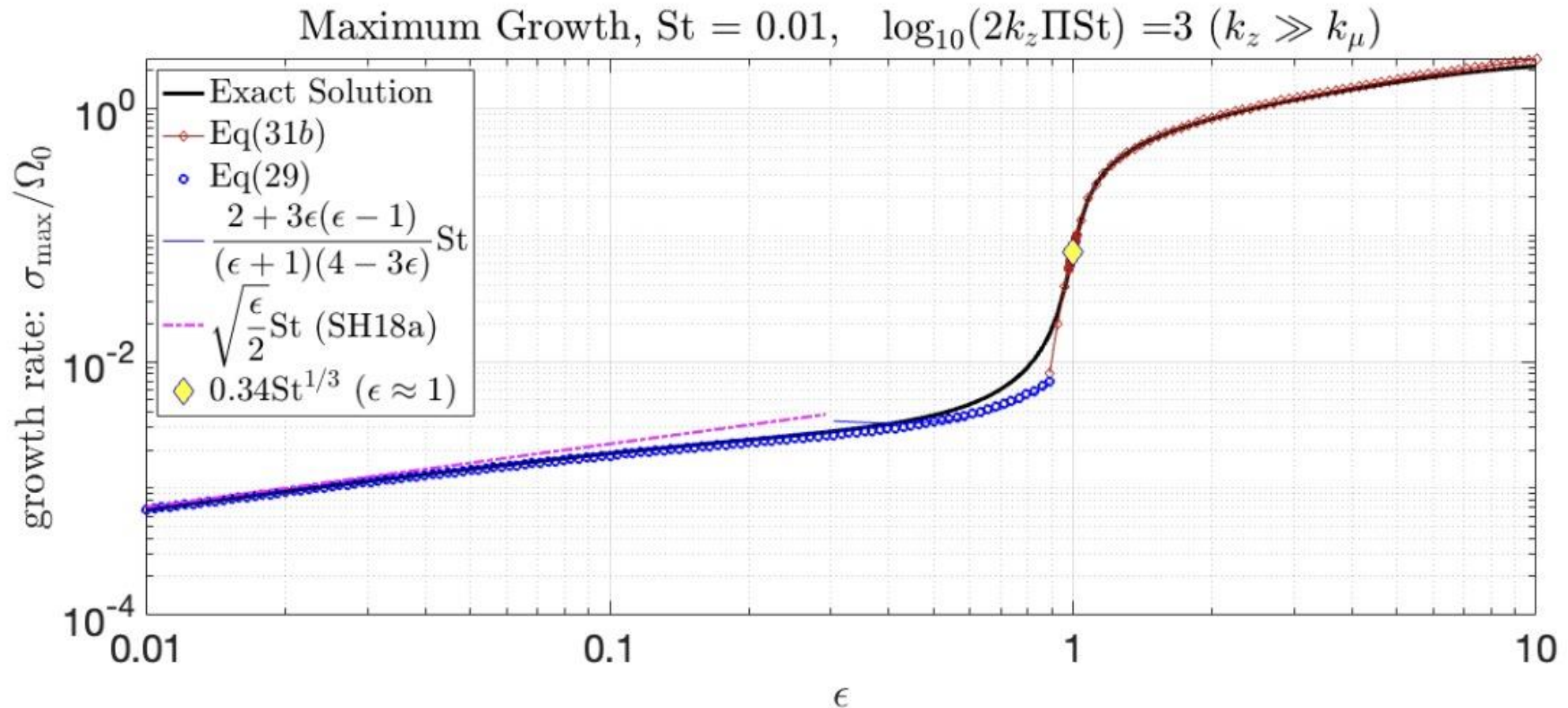
$$(xi) \quad \underline{\epsilon > 1 - St^{2/3}; \quad 2k_z \Pi St \gg (1 + \epsilon)^2:}$$

$$\begin{aligned} \sigma_{\max} &= \text{Im}(\omega_{\max}), \\ \omega_{\max} &= \frac{1}{2\kappa_m} \left(1 - i\kappa_m^3 - \sqrt{1 - 4\kappa_m^2(\epsilon - 1)St^{-2/3} - 6i\kappa_m^3} \right) St^{1/3} \\ &\quad + \left(\frac{1}{2\kappa_m} - i\frac{\kappa_m^2}{4} \right) (\epsilon - 1) St^{1/3} + \mathcal{O}(St^{2/3}), \\ k_{x,\max}\Pi &= \kappa_m \frac{(1 + \epsilon)^2}{2} St^{-4/3}, \end{aligned}$$

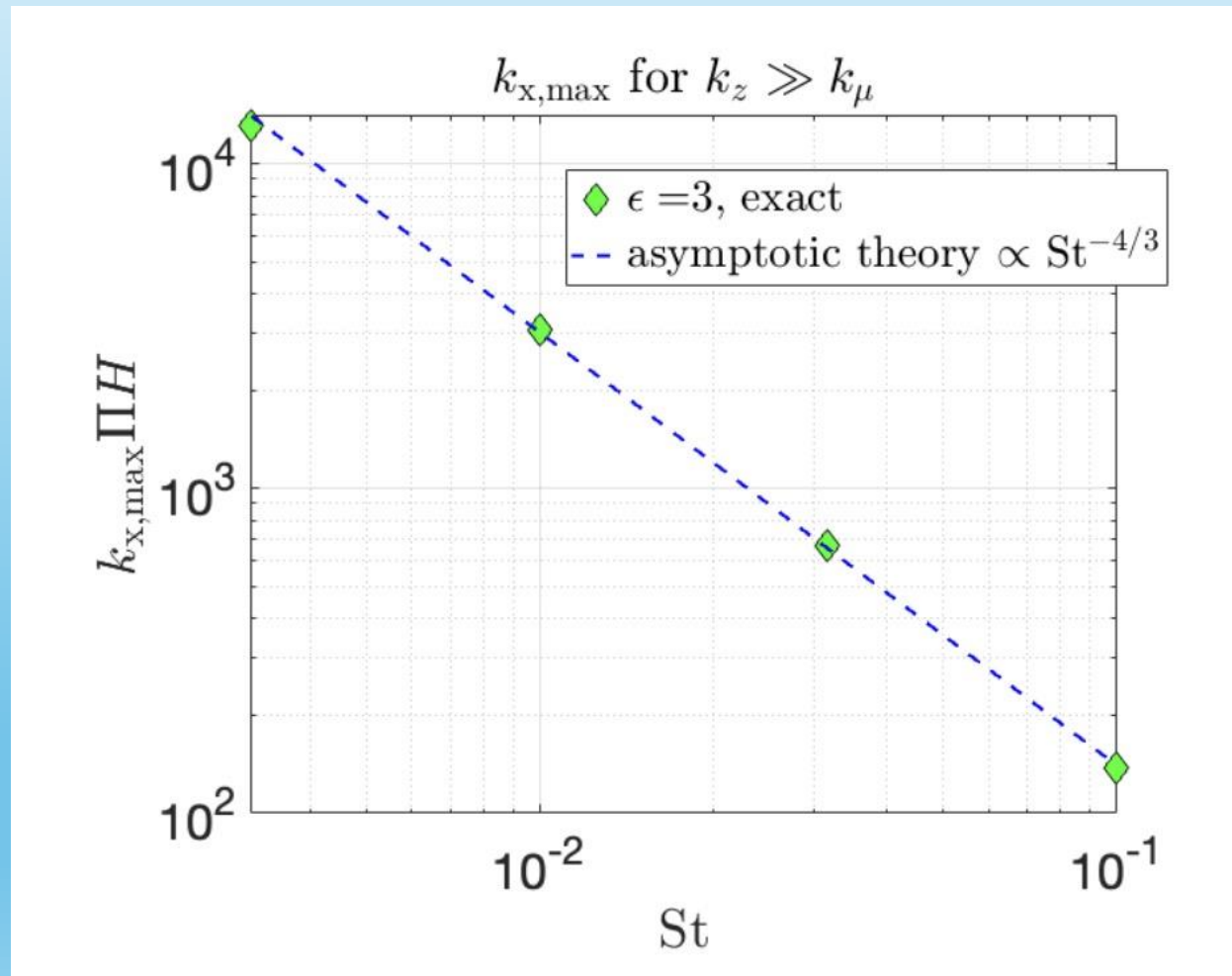
$$\begin{aligned} \sigma_{\max} &\approx 0.34 St^{1/3}, \\ k_{x,\max}\Pi &\approx 1.6 St^{-4/3}. \end{aligned}$$

$$(x) \quad \underline{\epsilon \approx 1; \quad 2k_z \Pi St \gg 4:}$$

Verification



Verification



Longwavelength Regime Results

$$k_z \ll k_\mu$$

$$(i) \quad \underline{\epsilon \ll 1; \quad 2k_z \Pi \text{St} \ll (1 + \epsilon)^2:}$$

$$\sigma_{\max} \approx \frac{1}{2} \sqrt{\epsilon/2} \text{St} + \mathcal{O}(\text{St}^2),$$
$$k_{x,\max} \Pi \approx \frac{1}{2} \sqrt{2k_z \Pi} \text{St}^{-1/2}.$$

Recovers Squire+Hopkins 18a in St small limit

$$(v) \quad \underline{\epsilon < 1 - \text{St}^{6/5}; \quad 2k_z \Pi \text{St} \ll (1 + \epsilon)^2:}$$

$$k_{x,\text{max}} \Pi \approx (1 + \epsilon) \sqrt{\frac{k_z \Pi}{2}} \text{St}^{-1/2}.$$

$$\sigma_{\text{max}} \approx \frac{\epsilon}{1 + \epsilon} \cdot \frac{\tilde{\omega}_0^2}{1 - 2\tilde{\omega}_0 - 3\tilde{\omega}_0^2} \text{St},$$

$$\tilde{\omega}_0 \equiv -1 + \sqrt{\frac{\epsilon}{2}} (1 + \epsilon) + (1 - \sqrt{2})\epsilon.$$

$$(viii) \quad \underline{|\epsilon - 1| \lesssim \text{St}^{6/5}; \quad 2k_z \Pi \text{St} \ll 4:}$$

$$\sigma_{\max} = \frac{(2k_z \Pi)^{4/5}}{4(1 + \epsilon)\chi_{\max}^{3/5}} \text{Im} \left[2 - i\chi_{\max} - 2 \left(1 - \frac{4(\epsilon - 1)\chi_{\max}^{4/5}}{(2k_z \Pi)^{2/5}} \text{St}^{-6/5} - 3i\chi_{\max} \right)^{1/2} \right] \text{St}^{7/5},$$

$$k_{x,\max} = \frac{1}{2}(1 + \epsilon) \left(\chi_{\max} (2k_z \Pi)^2 \right)^{1/5} \text{St}^{-4/5}.$$

number $\chi_{\max} = 0.545$.

$$\sigma_{\max} = 0.146 (2k_z \Pi)^{4/5} \text{St}^{7/5},$$

$$k_{x,\max} \Pi = \left(0.545 (2k_z \Pi)^2 \right)^{1/5} \text{St}^{-4/5}.$$

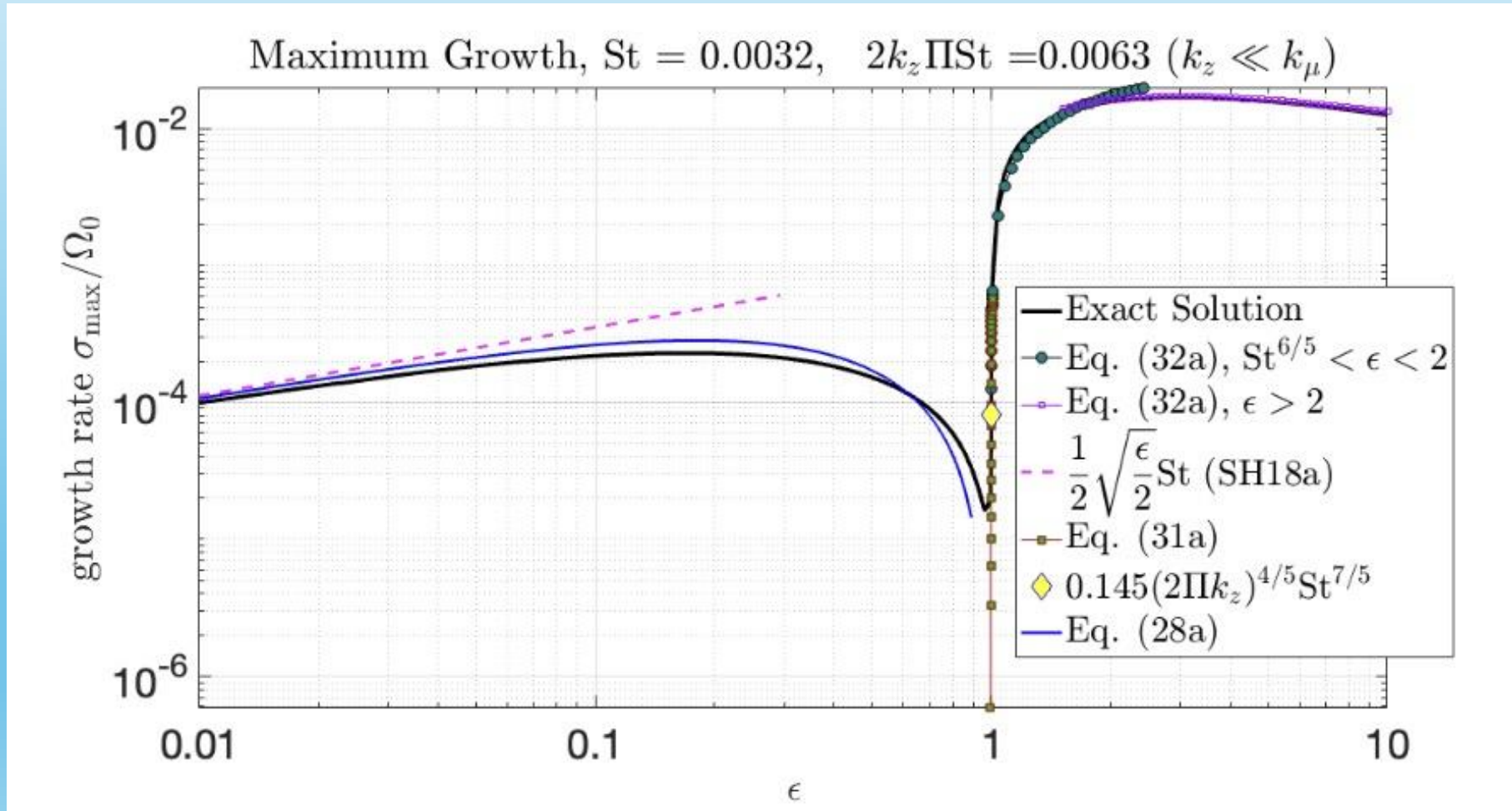
$$(vii) \quad \underline{\epsilon = 1; \quad 2k_z \Pi \text{St} \ll 4:}$$

$$(ix) \quad \underline{\epsilon > 1 + \text{St}^{6/5}; \quad 2k_z \Pi \text{St} \lesssim (1 + \epsilon)^2:}$$

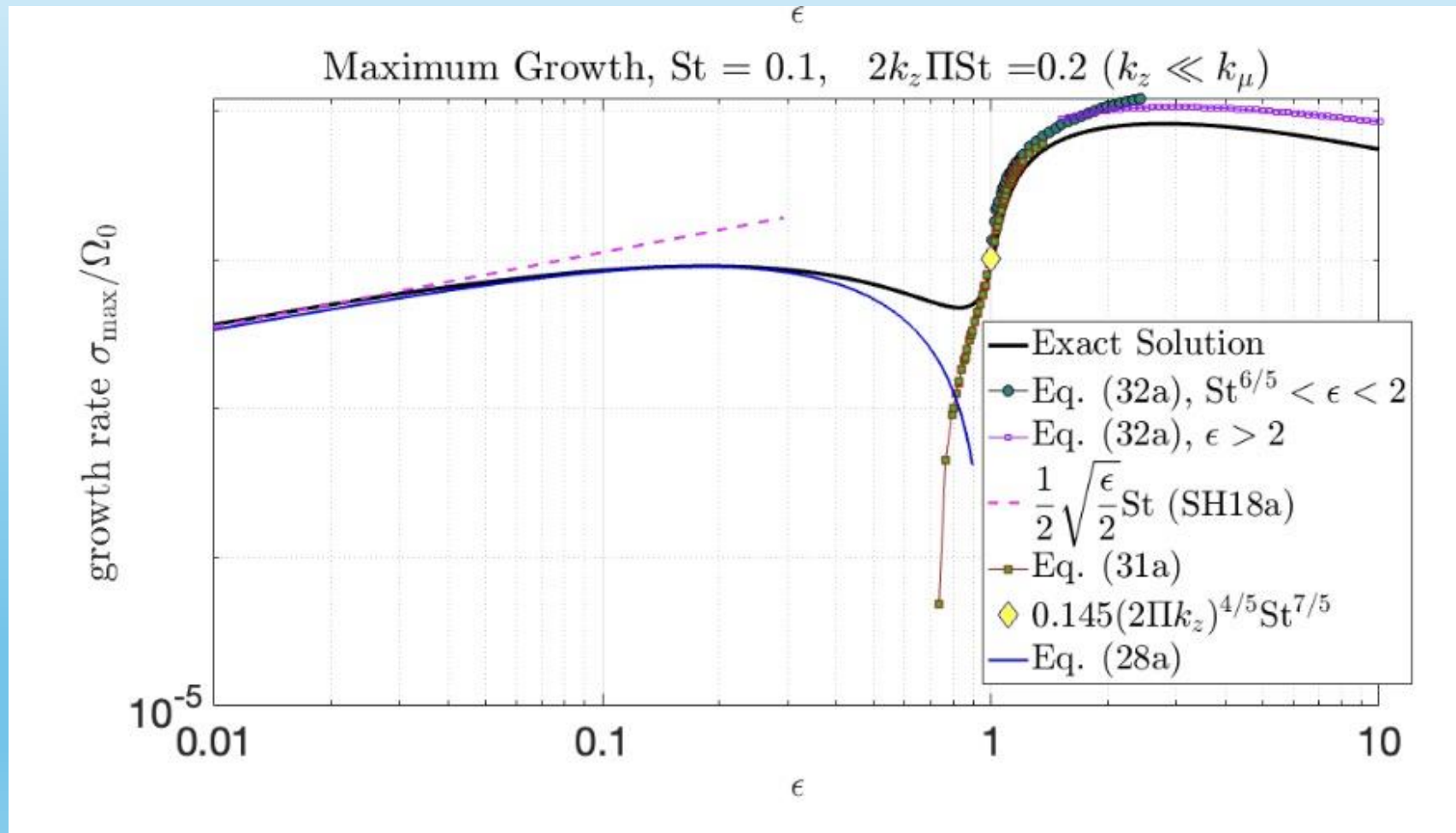
$$\sigma_{\max} \approx \begin{cases} 0.651 \sqrt{2\Pi k_z} \frac{(\epsilon - 1)^{3/4}}{1 + \epsilon} \text{St}^{1/2}, & \text{St}^{1/2} \ll \epsilon - 1 \ll 1, \\ 0.608 \sqrt{2\Pi k_z} \frac{(\epsilon - 1)^{1/2}}{1 + \epsilon} \text{St}^{1/2}, & 1 \ll \epsilon - 1 \ll \text{St}^{-1/2}. \end{cases}$$

$$k_{x,\max} \Pi \approx \begin{cases} 0.93 \frac{(1 + \epsilon)}{(\epsilon - 1)^{1/4}} \sqrt{\frac{k_z \Pi}{2}} \text{St}^{-1/2}, & \text{St}^{1/2} \ll \epsilon - 1 \ll 1, \\ 1.51 \frac{(1 + \epsilon)}{(\epsilon - 1)^{1/2}} \sqrt{\frac{k_z \Pi}{2}} \text{St}^{-1/2}, & 1 \ll \epsilon - 1 \ll \text{St}^{-1/2}. \end{cases}$$

Verification

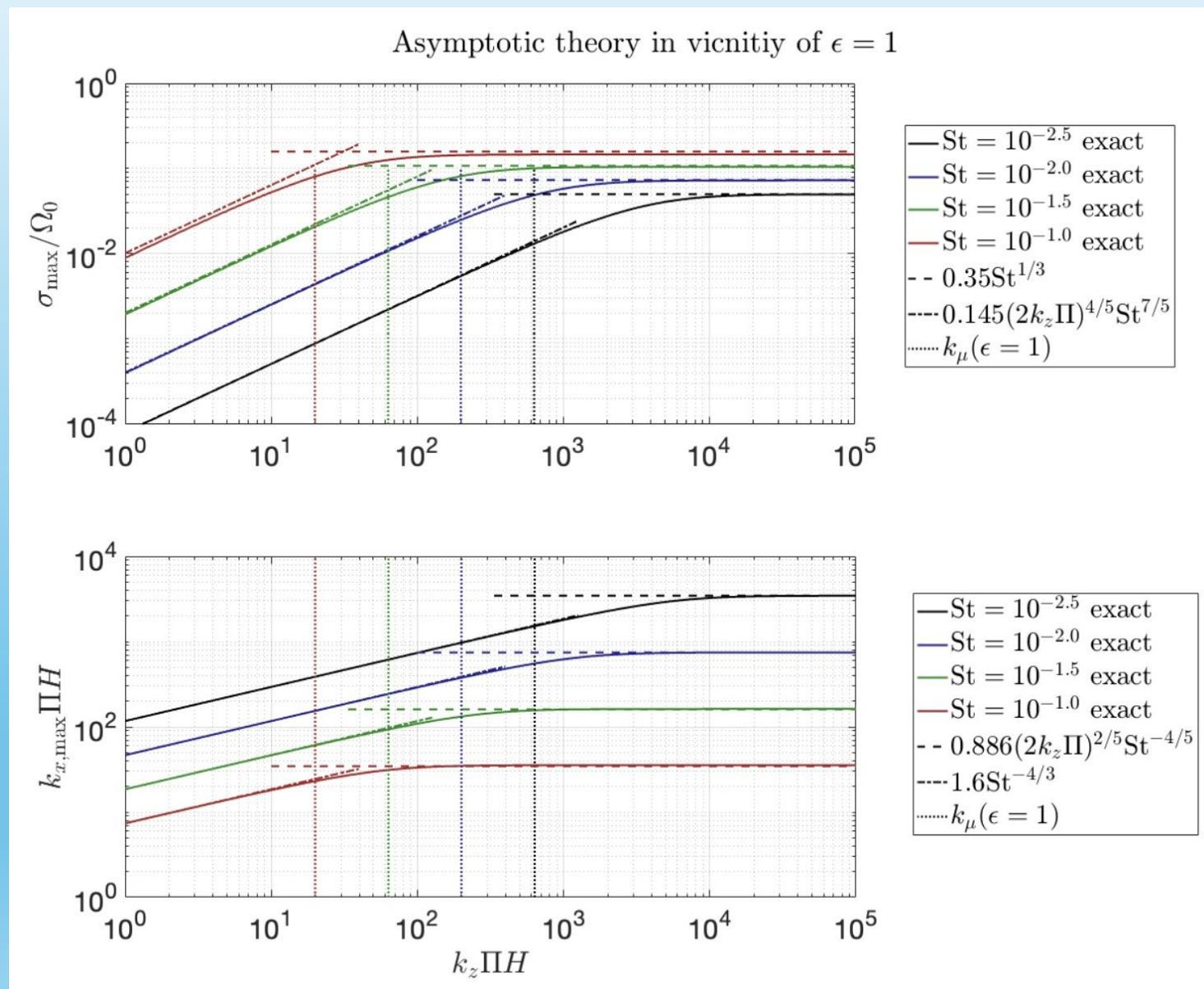


Verification



- Begins to fail around $St = 0.1$

Verification:
 $\epsilon=1$, general k_z



So what?

- We can use these estimates to analytically predict what happens to SI when in presence of other effects/timescales, e.g. turbulence.

➔ Understand findings of Umurhan+2020 and Chen+Lin 2020,
and of simulations in general

$$\sigma_{\text{net}} \approx \sigma_{\text{max}} - \sigma_{\text{turb}}$$

Li+Youdin 2021

How to model represent external turbulence

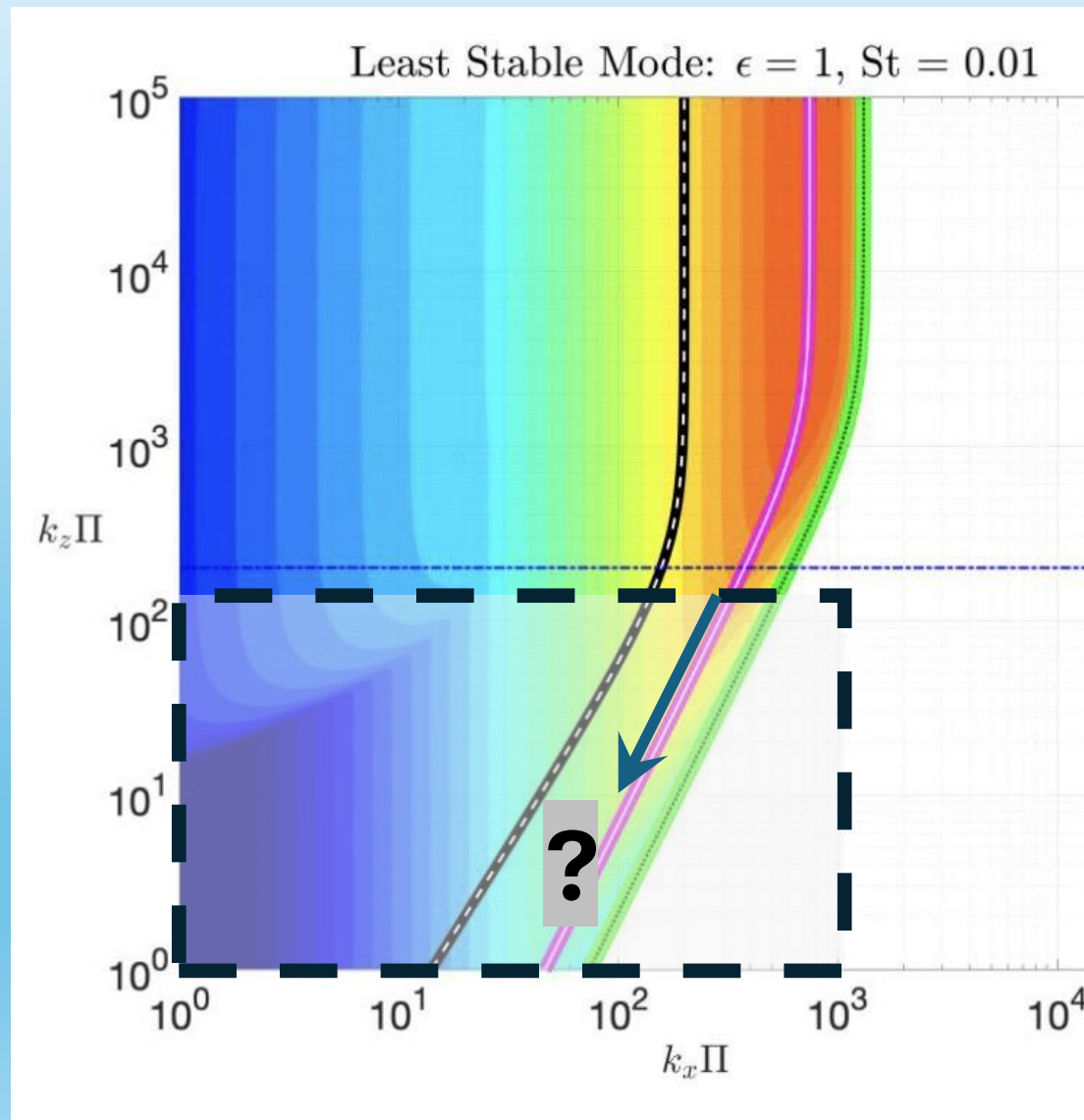
- Classically done as diffusion (mixing-length theory)

$$\sigma_{\text{turb}} = \alpha k^2$$

What happens at $\varepsilon=1$?

Suppose simulation
resolution only resolves the
longwavelength regime

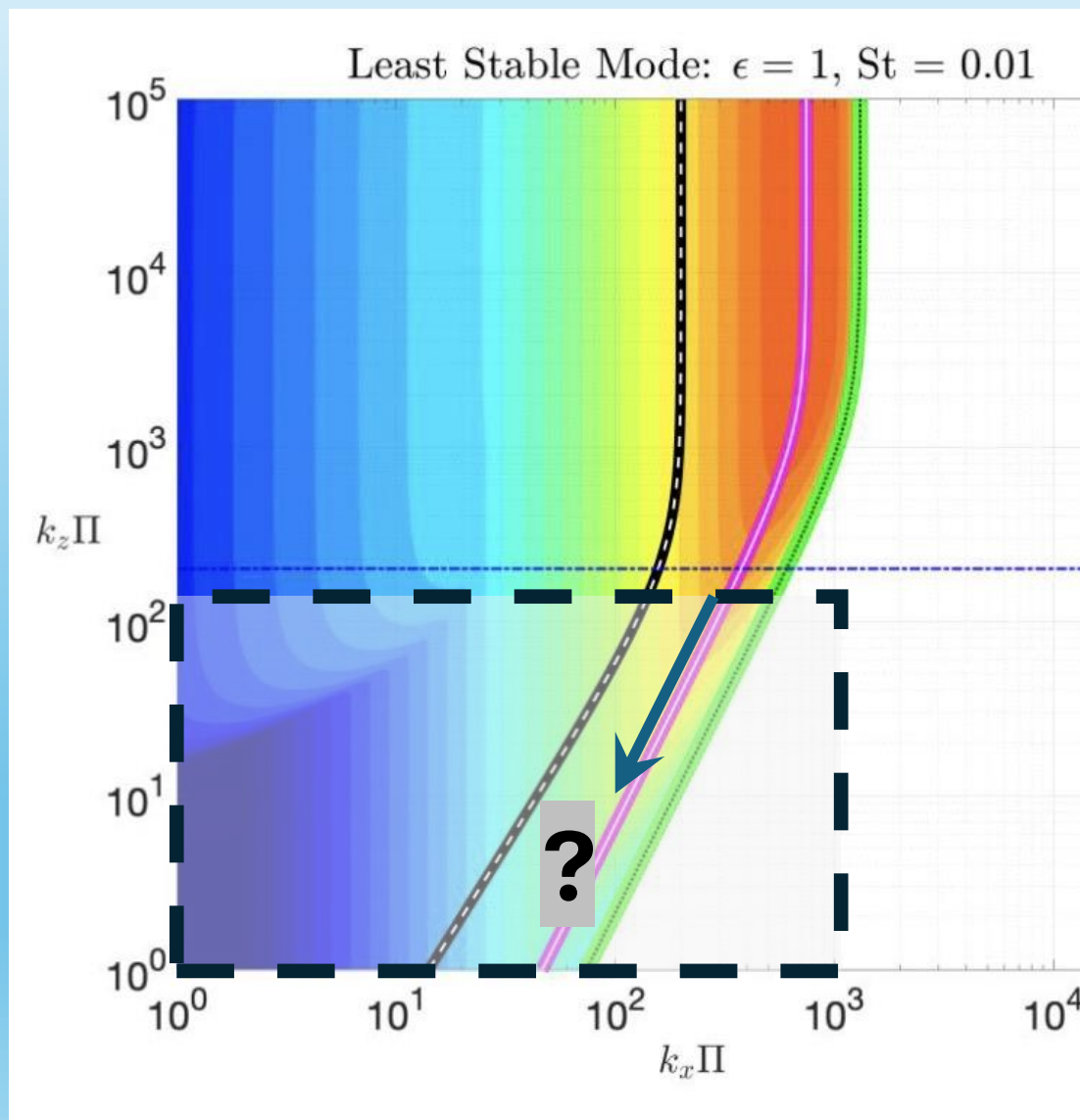
$$k_{x,\max} \gg k_z$$



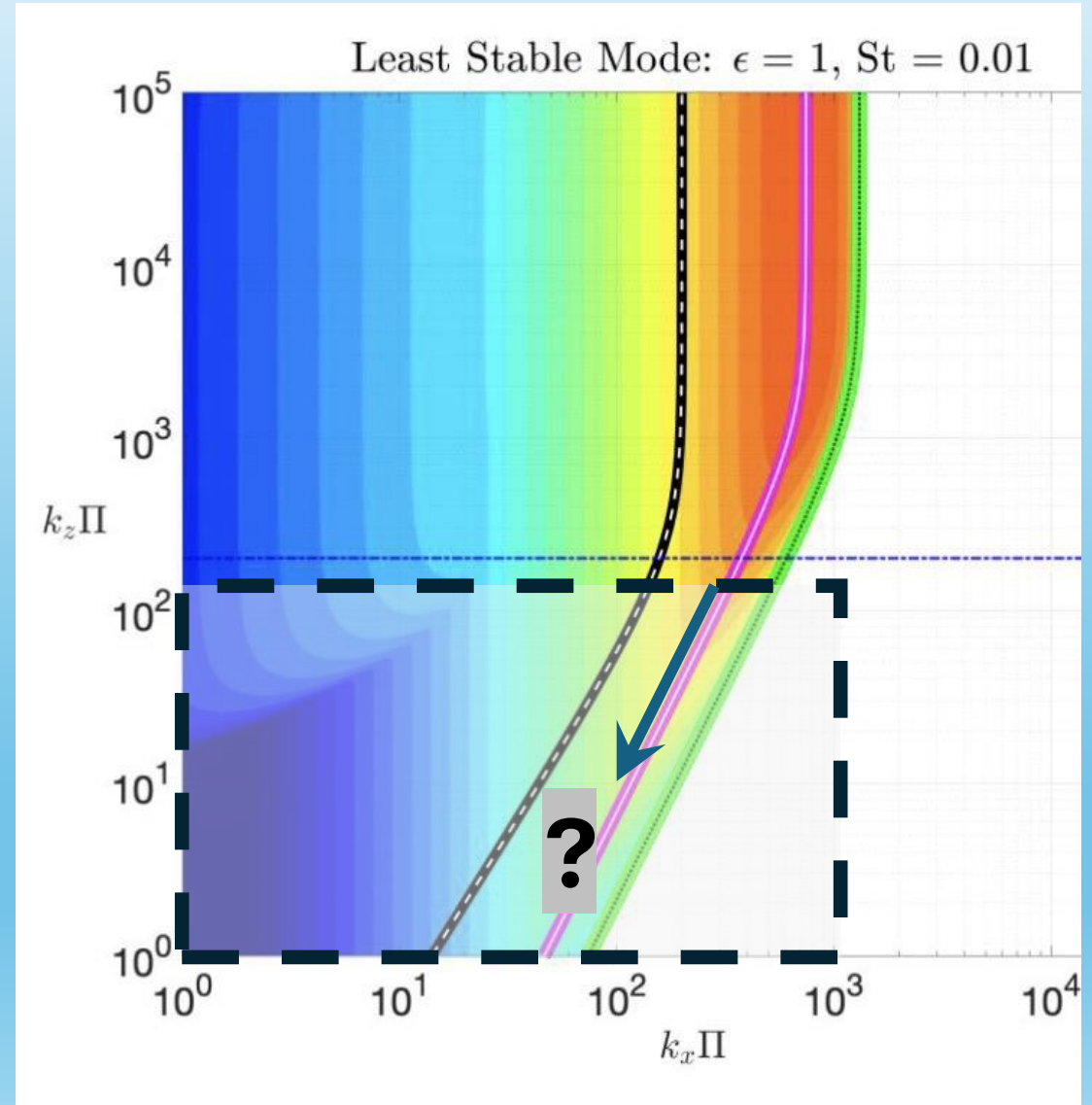
$$\sigma_{\text{net}} \approx \sigma_{\text{max}}(\epsilon = 1) - \alpha k^2$$

$$\sigma_{\text{max}} = 0.146 (2k_z \Pi)^{4/5} \text{St}^{7/5},$$

$$k_{x,\text{max}} \Pi = \left(0.545 (2k_z \Pi)^2 \right)^{1/5} \text{St}^{-4/5}.$$

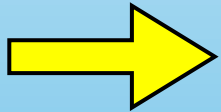


$$\sigma_{\text{net}} \approx 0.146 (2k_z \Pi)^{4/5} \text{St}^{7/5} \underbrace{\left(1 - \frac{\alpha}{\Pi^2 \text{St}^3}\right)}_{\equiv \Xi}$$

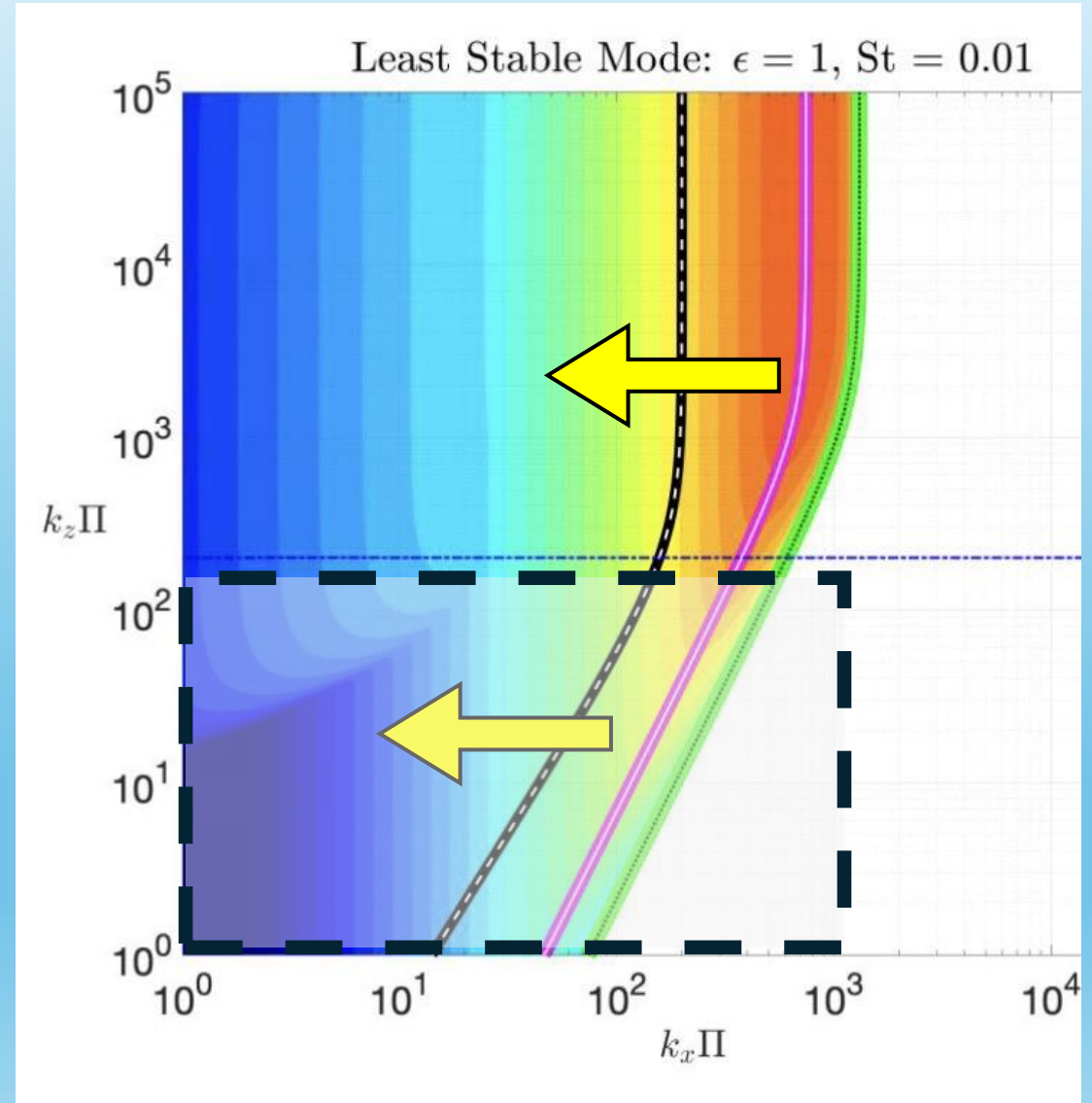


Does it mean there is no instability near $\epsilon=1$ with moderate turbulence?

Of course not!



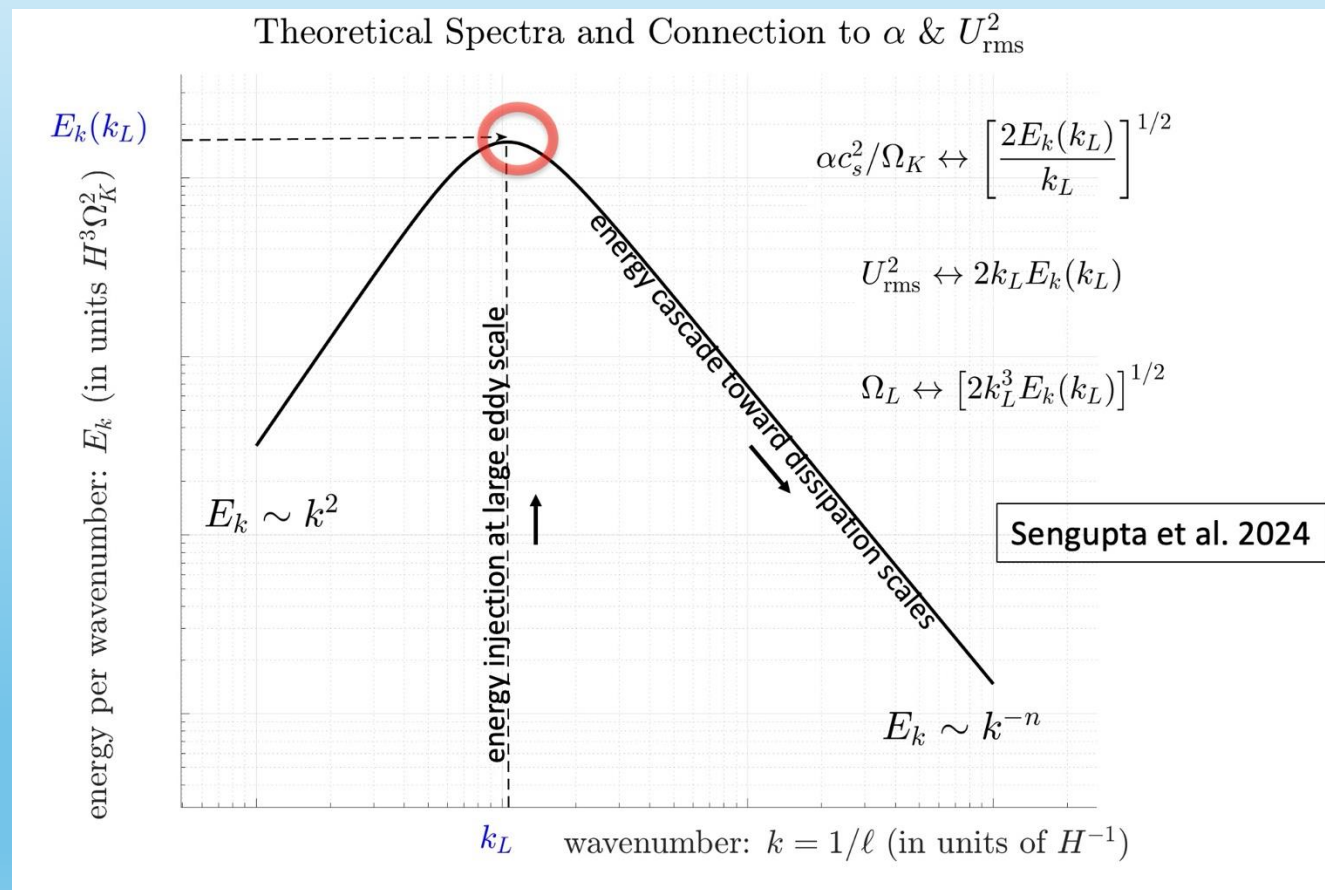
*Triggered modes are
not $k_{x,max}$
instead occur at
smaller k_x*



Summary

Supplemental Slides

But what if lengthscale of interest is in inertial range?



More general form

$$\sigma_{\text{turb}} = \begin{cases} \alpha k^2, & k < k_L \\ \Omega_L (k/k_L)^{n/2+3/2}, & k \geq k_L \end{cases}$$